

Arash Jalil Khabbazi, MS

🎓 PhD Student, Mechanical Engineering, Purdue University

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🎓 Google Scholar

Profile

I develop machine learning, optimization, and advanced control algorithms for intelligent buildings and energy systems. I bring hands-on experience from both research and industry.

Education

Purdue University, IN, United States 2023 – Present
PhD in Mechanical Engineering — **GPA:** 4.0/4.0
Minor: Computational Science and Engineering
Adviser: [Kevin J. Kircher](#)

University of British Columbia, BC, Canada 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.33/4.33 (94%)
Thesis: [Mixing gaseous hydrogen into natural gas distribution pipelines](#)
Adviser: [Sunny R. Li](#)

University of Tabriz, EA, Iran 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+
Thesis: *Thermodynamic analysis of Kalina cycle system 11*
Adviser: [S. Mohammad S. Mahmoudi](#)

Industry experience

Summer Intern, [Tesla](#) (Reno, NV, United States) 05/2025 – Present

- Developed supervised machine learning (ML) algorithms (SVR, Random Forest, XGBoost, stacking) for real-time quality assessment, eliminating destructive lab tests and achieving an R^2 above 0.8.
- Utilized MySQL for data extraction and management, performing feature selection analyses (Python, JMP) to identify high-signal inputs for ML model development.
- Contributed to deployment of an ML-based thermal imaging system for real-time defect detection and process optimization.
- Collaborated with Inverter Engineering and Advanced Manufacturing teams to align models with ongoing projects, with deployment strategies projected to save Tesla over \$1M annually in testing costs.

Student Researcher, [FortisBC](#) (Vancouver, BC, Canada) 09/2021 – 09/2023

- Supported FortisBC's technical assessments on hydrogen injection into natural gas pipelines and other renewable gas projects, including the UBC H₂Lab.
- Participated in monthly meetings with FortisBC engineers and research teams (materials, sensors, AI, safety) to review project findings and assess industry impacts.

Research experience

PhD Research Assistant, Purdue University 09/2023 – Present

- Supported commissioning of [Herrick Laboratories](#) as a testbed for advanced HVAC control, contributing to BAS retrofits (Niagara/Modbus) and developing occupant-facing dashboards for real-time comfort and energy feedback.
- Built ML models for occupancy detection using CO₂ sensor data with 98.3% accuracy and automated pipelines (Python, InfluxDB, Linux) for real-time energy management in smart HVAC control.
- Designed and simulated HVAC control strategies with MPC and DeePC (CVX, CVXPY), showing DeePC matched MPC performance without requiring a system model, while addressing energy efficiency and occupant comfort.
- Conducted a large-scale review of 100+ publications (200+ field tests) on MPC and RL for residential and commercial HVAC, leading to a published [review paper](#) in Applied Energy.
- Modeled thermodynamic performance of low-GWP refrigerants in heat pump systems for cold climates using EES, analyzing efficiency trade-offs to support sustainable HVAC design and decarbonization.

- Led research on hydrogen injection into natural gas pipelines using CFD, real-gas modeling (C), and CAD-based pipeline design.
- Analyzed over 1 TB of simulation data, resulting in a [journal paper](#), three conference presentations, and earning two conference awards.

Skills

Programming & Scripting: Python, C, MATLAB, R, Git, Markdown, HTML

ML & Data Science: TensorFlow, Keras, scikit-learn, XGBoost, NumPy, Pandas, SciPy, Matplotlib, Seaborn

Data Systems & Operating Environments: MySQL, InfluxDB, Grafana, Linux

Control & Communication Protocols: Modbus, IO-Link, MTConnect

Engineering & Simulation Tools: EES, ANSYS Workbench, OpenFOAM, Tecplot, SOLIDWORKS, CATIA, JMP

Publications

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review,” *Applied Energy*, 2025. [🔗](#)
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions,” *International Journal of Hydrogen Energy*, 2024. [🔗](#)

Conference proceedings

4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?,” *International High Performance Buildings Conference*, 2024, pp. 1–10. [🔗](#)
3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions,” *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6. [🔗](#)
2. **A. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid,” *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1–1. [🔗](#)
1. **A. Khabbazi**, R. Li, and J. Quinn, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines,” *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1–1. [🔗](#)

Honors & awards

Selected

ASHRAE Graduate Student Grant-in-Aid Award	ASHRAE, 2025
Best Paper Award at CSME 2023 Conference	CSME, 2023
Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium	CSME, 2022

Others

Purdue University Graduate School “Say It In 6” Finalist.	Purdue, 2024
UBC Graduate Scholarship.	UBC, 2022
UBC Dean’s Entrance Scholarship.	UBC, 2021

Selected courses

Thermal & Energy Systems: Distributed Energy Resources (ME597) — Analysis of Thermal Systems (ME518) — Advanced Thermodynamics (ME500) — Refrigeration Systems — Power Plants — Heat Transfer

Applied Mathematics & Data Science: Applied Machine Learning (ENGR418) — Statistical Methods (STAT511) — Advanced Mathematics I (MA527) — Numerical Computations

Automation, Control & IoT: Convex Optimization (AAE561) — Industrial IoT Implementation (ME597)

Fluids & Simulation: Computational Fluid Dynamics (CFD) — Fundamentals of CFD — Multiphase Flows — Turbulence